

Measuring Case-Base Consistency with Monotonicity Metrics

Wijnand van Woerkom

Max Planck Institute for Comparative and International Private Law
Hamburg, Germany

Abstract

This paper connects precedential constraint in formal legal theory with monotonicity constraints in data science and machine learning. We show that Horty’s result model of case-based reasoning, including its dimension-based variants, is a special case of monotonicity constraints on a dataset of observations. This connection allows case-base consistency to be analyzed using monotonicity metrics. We review and systematize such metrics from the machine learning literature, give graph-theoretic and regression-based interpretations, and derive relations and bounds among commonly used measures. We then apply these metrics to two legal-domain datasets—COMPAS recidivism risk scores and Pennsylvania bail decisions—and find that different metrics capture different aspects of case-base consistency and can be used to identify highly inconsistent cases.

CCS Concepts

• **Applied computing** → Law; • **Computing methodologies** → **Knowledge representation and reasoning**; *Machine learning*; • **Mathematics of computing** → Graph theory.

Keywords

Precedential constraint, Case-base consistency, Monotonicity metrics, Monotonic regression, Criminal justice decision-making

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1 Introduction

Case-based reasoning is an essential part of legal reasoning, especially in common law jurisdictions. Work on formal modelling of case-based reasoning in AI & law dates back to the HYPO and CATO models [1, 33]. These works inspired Horty to develop his formalization of the *result model* of a *fortiori* precedential constraint in [17]. This model succinctly formalizes the idea that when a new case differs from a precedent only in ways that make it a stronger case for the decision in that precedent, then that same decision should be reached in the new case. Subsequently developed variations of this model use more expressive knowledge representations [18, 19], as part of an effort to develop a general account of common law [20].

At the same time, the result model has increasingly been used in recent years in data-driven AI contexts. In [31], a method is

proposed for explaining data-driven decisions, consisting of an argumentation framework that is built on top of the result model. Work on this method has continued in [29, 30]. In fact, the result model has itself been used as an interpretable data-driven decision system. It was modified to operate with uncertain information in [28], and subsequently integrated into a system for reporting online fraud [27] which is actively used by the Dutch national police. Similarly, a probabilistic extension was recently developed and shown in an experiment to perform on par with other machine learning methods, while remaining interpretable [26]. Lastly, building on [31], [39] argues that variants of the result model can be used as a simple normative fairness principle, and showcased this by a case study of the COMPAS recidivism prediction system.

In this article we establish a connection between work on the result model and that on *monotonicity constraints* in computer science. Such constraints arise in settings where outputs are produced on the basis of inputs, and state that an increase in an input variable, *ceteris paribus*, should lead to an increase in the output variable [38]. As [38] puts it, “this concept is both intuitive and ubiquitous.” Indeed, monotonicity constraints have been studied and applied in a wide variety of research areas. In particular, in many areas of machine learning and statistics the goal is to train a model that satisfies monotonicity constraints. The literature on monotonic regression dates back to at least the 1950s; see, e.g., the references in [22]. Similarly, monotonic versions of many well-known algorithms have been developed, including neural networks, support vector machines, and decision trees; see [4] for a recent overview. Furthermore, monotonicity constraints have been studied since at least the early 2000s in *property testing*, a subfield of theoretical computer science; see [6] and the references therein. Applications of these methods span diverse fields, including finance, medicine, and law [3, 7, 8, 38].

Our first contribution is to show that the result model, including its dimension-based variants, is a special case of monotonicity constraints. We develop this connection in detail in Section 2. This identification lets case-base consistency be measured as dataset monotonicity, for which the literature on monotonicity constraints offers a range of metrics and methods of assessment [9, 15, 21, 24, 25, 41]. Our second contribution is to review and organize these metrics, provide graph-theoretic and regression-based interpretations, and derive relations and bounds among commonly used measures. We present these results in Section 3. Our third contribution is to apply the metrics to two legal-domain datasets in Section 4: COMPAS recidivism risk scores [2, 34], and Pennsylvania bail decisions [42]. These analyses show that different measures capture different aspects of case-base consistency and can help localize particularly inconsistent cases in practice within these legal settings. We conclude with some directions for future research in Section 5.

2 Monotonicity and Precedential Constraints

In this section, we introduce the basic setup of monotonicity constraints on datasets, and then show how this setup captures the result model of precedential constraint. Before proceeding, we first fix some notation. Let A be a set of elements. The cardinality of A is denoted $|A|$. For a number $n \in \mathbb{N}$ we let $A^n = \{(a_1, \dots, a_n) \mid a_1, \dots, a_n \in A\}$ denote the set of all ordered n -tuples of elements from A , and $A^{\neq} = \{(a_1, \dots, a_n) \in A^n \mid i \neq j \text{ implies } a_i \neq a_j\}$ the set of ordered distinct n -tuples from A , and $\binom{A}{n} = \{t \subseteq A \mid |t| = n\}$ the set of all unordered distinct n -tuples from A .

2.1 Monotonicity Constraints on a Dataset

Monotonicity constraints are phrased with respect to a finite set of observations $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where $n \in \mathbb{N}$ is a natural number and where every observation $(\mathbf{x}_i, y_i)$ is a pair consisting of a d -dimensional input (vector) $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,d}) \in \mathbb{R}^d$ and an output $y_i \in \mathbb{R}$. Writing $N = \{1, \dots, n\}$, it can be convenient to think of such a dataset as specified by functions $\mathbf{x} : N \rightarrow \mathbb{R}^d$ and $y : N \rightarrow \mathbb{R}$, and to conflate the elements of N with the observations (i.e., to refer to $i \in N$ as “the observation,” rather than the pair $(\mathbf{x}_i, y_i)$ that it indexes), and we may do so henceforth. In the literature on monotonic classification the y_i values fall in a finite set, often denoted $\mathcal{L}$, and are referred to as *labels*. In this context, a function $y : N \rightarrow \mathcal{L}$ is called a *labeling*. This terminology has become sufficiently common that even in the context of regression (when the outputs live in $\mathbb{R}$), the y_i values may still be referred to as labels, and $y : N \rightarrow \mathbb{R}$ as a labeling.

The fundamental additional *monotonicity assumption* around which this and similar work revolves is that an increase in an input variable, ceteris paribus, leads to an increase in the output variable [38]. To formalize this we use the *product order* $\leq$ on the set of inputs, defined by $\mathbf{x}_i \leq \mathbf{x}_j$ iff $x_{i,k} \leq x_{j,k}$ for $1 \leq k \leq d$. Note that this relation is not *total*: given two inputs $\mathbf{x}_i$ and $\mathbf{x}_j$ it is not necessarily the case that either $\mathbf{x}_i \leq \mathbf{x}_j$ or $\mathbf{x}_j \leq \mathbf{x}_i$ holds. If the product order holds in either direction then we say that the pair of inputs is *comparable*, and *incomparable* otherwise. We now have:

Definition 2.1. Let $(\mathbf{x}, y)$ be a dataset of n observations, then a pair of observations $(i, j) \in N^2$ is *monotone* when

$$\mathbf{x}_i \leq \mathbf{x}_j \text{ implies } y_i \leq y_j. \quad (\text{M})$$

If (i, j) is not monotone, i.e. when $\mathbf{x}_i \leq \mathbf{x}_j$ and $y_i > y_j$, then we say it is *nonmonotone*, or that it is a (*monotonicity*) *violation*. A dataset is *monotone* when every pair of observations $(i, j) \in N^2$ is monotone, and *nonmonotone* otherwise.

We let Definition 2.1 quantify over N^2 instead of $N^{\neq}$ because a pair (i, i) always satisfies (M). However, we should emphasize that different observations can still have the same inputs and outputs: $i \neq j$ does not necessarily imply $\mathbf{x}_i \neq \mathbf{x}_j$ or $y_i \neq y_j$.

Note that Definition 2.1 constitutes a *directed* notion of monotonicity: to say that a pair of observations (i, j) is monotone is different from saying that (j, i) is monotone. However, nonmonotonicity can occur in at most one direction at a time:

Lemma 2.2. If $(i, j) \in N^2$ is nonmonotone then (j, i) is monotone.

Proof. If (i, j) and (j, i) were both nonmonotone then $y_i > y_j$ and $y_j > y_i$, which is a contradiction. $\square$

A central topic of the present work is the quantification of monotonicity. One way of doing this is by considering the relative frequency of nonmonotonic pairs of observations. Lemma 2.2 shows that this frequency will be at most one-half, which suggests that it is better measured as a property of *unordered* pairs. For this reason, we will also consider nonmonotonicity as an undirected notion:

Definition 2.3. An unordered pair of observations $\{i, j\} \in \binom{N}{2}$ is *nonmonotone*, when either (i, j) is nonmonotone or (j, i) is nonmonotone; $\{i, j\}$ is *monotone* when it is not nonmonotone.

Example 2.4. To illustrate these definitions, and in anticipation of our data analyses in Section 4, we consider an example domain on recidivism risk assessments. Some relevant factors that may be taken into account for such a decision are the defendant’s age, whether they have a history of criminal involvement, and their level of education [11]. History of criminal involvement has a monotonic influence on risk: the more prior offenses a person has committed, the higher the risk of a new offense. Age and level of education are similarly correlated, but in a negative way: higher values for these variables suggest a lower risk. This may nevertheless be incorporated in the framework simply by recording the negative of the actual values. In turn, we could consider a decision about bail to be based on recidivism and flight risk assessments, where both have a monotonic influence on the amount of bail that is assigned.

To formalize this, we consider these inputs as triples consisting of (the negative of) age, the number of priors, and (the negative of) a 1 or 0 representing a completion of high school education. Suppose we have a set $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), (\mathbf{x}_3, y_3)\}$ of observations:

$$\begin{aligned} \mathbf{x}_1 &= (-25, 7, 0), & y_1 &= 7, \\ \mathbf{x}_2 &= (-30, 5, -1), & y_2 &= 5, \\ \mathbf{x}_3 &= (-20, 6, -1), & y_3 &= 3. \end{aligned}$$

Now we have $\mathbf{x}_2 \leq \mathbf{x}_1$ and $\mathbf{x}_2 \leq \mathbf{x}_3$, whereas observations 1 and 3 are incomparable. The pair $(2, 1)$ is monotone because $y_2 = 5 \leq y_1 = 7$, but $(2, 3)$ is nonmonotone because $y_2 = 5 > y_3 = 3$. The pair $(1, 3)$ is trivially monotone because $\mathbf{x}_1 \not\leq \mathbf{x}_3$. In terms of unordered pairs, $\{1, 2\}$ and $\{1, 3\}$ are monotone whereas $\{2, 3\}$ is nonmonotone.

2.2 The Result Model of Precedential Constraint

We now show that Horty’s result model [17] and its subsequent dimension-based expansions [19, 40] are instances of monotonicity constraints. We treat the former in Section 2.2.1 and the latter in Section 2.2.2.

2.2.1 The Factor-Based Model. The factor-based result model represents cases as sets of *factors*: abstract legal fact patterns that are assumed to consistently favor either the plaintiff or the defendant of a dispute. Formally these are given as disjoint sets $F = F_\pi \cup F_\delta$, where F_π contains the pro-plaintiff factors and F_δ the pro-defendant factors. A *fact situation* is a subset $X \subseteq F$, and a *case* C is a pair $C = (X, s)$ where X is a fact situation and $s \in \{\pi, \delta\}$ is called the *outcome* of C . A *case base* is a set of cases $\mathcal{C}$. We write $\bar{s}$ to denote the outcome opposite to $s \in \{\pi, \delta\}$, so $\bar{\pi} = \delta$ and $\bar{\delta} = \pi$. For a fact situation X we let $X_s = X \cap F_s$ denote the set of pro- s factors in X .

Horty then defines a *strength order* $\leq_s$ on fact situations $X, Y \subseteq F$:

$$X \leq_s Y \quad \text{iff} \quad X_s \subseteq Y_s \text{ and } Y_{\bar{s}} \subseteq X_{\bar{s}}. \quad (1)$$

Intuitively, $X \leq_s Y$ means that there is more support for s in Y relative to X , or conversely, that there is more support for $\bar{s}$ in X relative to Y . This gives rise to the notion of constraint: when (X, s) is a case that was decided for s , and Y is a new fact situation such that $X \leq_s Y$, then the decision in Y is *constrained* (or, *forced*) for s . Inconsistency is then defined as a violation of this constraint.

Definition 2.5. A pair of cases $C = (X, s)$ and $D = (Y, t)$ are *inconsistent* when $X \leq_s Y$ and $t = \bar{s}$, and *consistent* otherwise. A case base C is *inconsistent* if it contains a pair of inconsistent cases, and *consistent* otherwise.

The point is now that Definition 2.5 can be seen as an instantiation of Definition 2.1, where all the inputs and outputs are two-valued. To see this, let d denote the number of factors in $F = \{f_1, \dots, f_d\}$. A fact situation X can now be seen as a vector $\mathbf{x}_X \in \mathbb{R}^d$ where the k th coordinate $\mathbf{x}_{X,k}$ is defined as:

$$\mathbf{x}_{X,k} = \begin{cases} 1 & \text{if } f_k \in X_\pi, \\ 0 & \text{if } f_k \notin X, \\ -1 & \text{if } f_k \in X_\delta. \end{cases} \quad (2)$$

Note that now $X \leq_\pi Y$ if and only if $\mathbf{x}_X \leq \mathbf{x}_Y$.

Similarly, we can encode the case outcomes π and δ simply as the numbers 1 and 0, respectively, so that a case base C can be interpreted as a set of observations in the sense of Section 2.1. Indeed, using this encoding and the fact of an equivalence between $X \leq_\pi Y$ and $\mathbf{x}_X \leq \mathbf{x}_Y$ we can see that the notions of consistency and monotonicity are aligned. Rather than giving a complete proof of this fact, we will simply showcase it through our running example.

Example 2.6. Suppose we make a binary assessment of high or low recidivism risk. We associate the high risk outcome with the plaintiff π , and low risk with the defendant δ . We could have three factors upon which the decisions are made, where superscripts indicate the preference of the factors towards the outcomes:

- f_1^δ : The defendant is older than 25,
- f_2^π : The defendant has a criminal record,
- f_3^δ : The defendant has finished high school.

Consider fact situations $X = \{f_2^\pi, f_3^\delta\}$ and $Y = \{f_1^\delta, f_2^\pi, f_3^\delta\}$. Now $X_\delta = \{f_3^\delta\} \subseteq Y_\delta = \{f_1^\delta, f_3^\delta\}$ and $Y_\pi = \{f_2^\pi\} \subseteq X_\pi = \{f_2^\pi\}$, which means $X \leq_\delta Y$. Therefore, the cases (X, δ) and (Y, π) are inconsistent according to Definition 2.5.

Applying the mapping of (2) to these fact situations we get $\mathbf{x}_X = (0, 1, -1)$ and $\mathbf{x}_Y = (-1, 1, -1)$, and indeed we see that $\mathbf{x}_Y \leq \mathbf{x}_X$. Representing an outcome for the plaintiff or defendant as a 1 or a 0 respectively, we can translate the cases (X, δ) and (Y, π) to observations $i_X = (\mathbf{x}_X, 0)$ and $i_Y = (\mathbf{x}_Y, 1)$, and verify that the pair (i_Y, i_X) is nonmonotone in the sense of Definition 2.1.

2.2.2 The Dimension-Based Models. In [19] Horty extends his result model to incorporate nonbinary factors which are called *dimensions* in the AI & law literature. Formally, a *dimension* is a set d with a partial order $\leq_d$ on d . Given values $v, w \in d$, the relation $v \leq_d w$ indicates that value w argues more strongly for a decision for the plaintiff relative to v , while conversely v argues more strongly for a decision for the defendant. We will subsequently omit the d subscripts. The dimension-based result model assumes that there is

a finite set of dimensions D that describe the domain at hand. A fact situation is given as a choice function on D , i.e. as a function $X : D \rightarrow \bigcup D$ satisfying $X(d) \in d$ for all $d \in D$. Outcomes and cases are modeled in the same way as in the factor-based setting, and the strength order $\leq$ is just the product order on dimensions:

$$X \leq Y \quad \text{iff} \quad X(d) \leq Y(d) \text{ for all } d \in D. \quad (3)$$

The strength order $X \leq Y$ states that Y has at least as much support for the plaintiff as X . The principle of constraint states that if (X, π) and $X \leq Y$, then Y should also be decided for the plaintiff. A dual of this principle applies to precedents for the defendant.

The dimensional-result model was expanded in [40] to incorporate hierarchies of dimensions. This allows cases to contain multiple dimension-valued decisions, rather than just a single plaintiff/defendant outcome. We will consider the special case of *flat* hierarchies, which contains just a single dimension-valued decision. Again there is a set of dimensions D , and in addition there is now an outcome dimension d_s . A fact situation is again a choice function on D , and a case is now a choice function on $D \cup \{d_s\}$. The strength order $\leq$ is as defined in (3), and the principle of constraint states that the values of cases on the outcome dimension should reflect the strength order; in sum:

$$\text{If } X(d) \leq Y(d) \text{ for all } d \in D, \text{ then } X(d_s) \leq Y(d_s). \quad (4)$$

The connection between (4) and (M) is clear when we interpret the values of the fact situations on D as the input variables $\mathbf{x}$, and the values on the outcome dimension d_s as the output variables y . For this correspondence to be exact, however, the relation in (M) should be an arbitrary partial order on the input space rather than specifically the coordinatewise product order. This more general order-theoretic formulation also appears in the literature; see for example [14, 22, 35, 36]. Secondly, the full model of [40] is more expressive than (M) due to the hierarchical structures that it allows. On the other hand, the way it defines constraint for these hierarchies is simply by recursively applying the monotonicity constraint (M).

3 Measuring Monotonicity

According to Definition 2.1 a dataset is either monotone or it is not, with no other options in between. This section is concerned with various ways of quantifying the amount of (non)monotonicity in a dataset, so that we can make more fine-grained, nonbinary assessments about dataset monotonicity; for example, one can consider the ratio of nonmonotone pairs to the total number of pairs of observations.

As shown in the previous section, dataset monotonicity coincides with the notion of case-base consistency of the result model. This means that we can use the methods that have been developed in computer science to analyze dataset monotonicity to reason about case-base consistency.

We start in Section 3.1 by looking at possible causes of dataset nonmonotonicity in order to provide some intuition. Then, in Sections 3.2 and 3.3 respectively, we recall some formal definitions from the literature on graph theory and monotonic regression. We will need these in Section 3.4, where we review commonly used monotonicity metrics, and propose a few new ones. Lastly, these metrics are then related to one another in Section 3.5, in order to guide our subsequent data analyses in Section 4.

3.1 Sources of Nonmonotonicity

In a way, our assumptions seem to be at odds: on the one hand we expect that the input variables have a monotonic influence on the output variable, while on the other hand we expect to observe non-monotone datasets and thus require fine-grained nonmonotonicity measures. As a matter of fact, there are several reasons why one could observe nonmonotone datasets in a setup that satisfies the monotonicity assumption, and we will now outline a few of them.

Measurement noise. It often happens that the observations in the dataset contain measurement noise, and this noise may cause nonmonotonicity that would not be present otherwise [5, 9, 32, 37]. This can hinder algorithms that are developed to train specifically on monotone datasets, and algorithms have been proposed in [5] to remove this kind of noise from the dataset.

As a normative principle. Depending on the domain, monotonicity constraints may be either descriptive or normative. In housing data, monotonicity is typically a modeling assumption about the underlying relationship. In other settings, such as recidivism risk assessment, monotonicity is often imposed normatively: we may require a decision rule to satisfy it even when the observed labels do not. This applies both to learned systems and to human decisions. As noted in [37], “due to the subjective nature of the labeling process [...], some datasets may not be fully monotonic.”

Decisionmaker multiplicity. When the labels are produced by not one but multiple decisionmakers, then the resulting dataset could be nonmonotone even if the decisionmakers are all independently monotone. For example, we might have two persons with the same relevant attributes who are judged to receive different amounts of bail by different, individually monotone judges.

Input incompleteness. It may happen that the inputs in the dataset are only a subset of the real set of inputs upon which the outputs are based. In this case, the labels might be produced in a monotonic manner, but seem nonmonotone when regarded as based on the restricted subset. This argument is also made in [39].

3.2 The Violation Graph

We now briefly recall some notions from graph theory. An undirected graph, or simply a graph, is a pair $G = (V, E)$ consisting of a set V of *vertices* (also called *nodes*) together with a set $E \subseteq \binom{V}{2}$ of unordered pairs of vertices called *edges*. A directed graph, also called a *digraph*, is a pair $D = (V, A)$ consisting of a set of vertices V together with a set $A \subseteq V^2$ of ordered pairs of distinct vertices called *arcs*. If a (di)graph has a set of vertices V then we speak of a (di)graph on V . A graph $G = (V, E)$ is a *subgraph* of another graph $G' = (V', E')$ when $V \subseteq V'$ and $E \subseteq E'$. We may similarly speak of subdigraphs. Any directed graph $D = (V, A)$ gives rise to an undirected graph $U(D) = (V, E)$ by forgetting the direction of the arcs of D : $E = \{\{v, v'\} \mid (v, v') \in A\}$. This is called the *underlying graph* of D . Two vertices in a graph are called *adjacent* when they are connected by an edge. For a vertex v in a graph $G = (V, E)$ we write $\deg(v)$ for the number of vertices adjacent to v , and $\deg(G)$ for the average degree of vertices in G . Note that $\deg(G) = 2|E|/|V|$. The *density* $\rho(G)$ of G is its number of edges divided by the total number of possible edges: $\rho(G) = |E|/\binom{|V|}{2}$.

Definition 3.1. Let $(\mathbf{x}, y)$ be a dataset of n observations, and $N = \{1, \dots, n\}$ the set of integers up to n . The *comparability digraph* on N is given by $D_{\leq} = (N, A_{\leq})$, where $A_{\leq} = \{(i, j) \in N^2 \mid \mathbf{x}_i \leq \mathbf{x}_j\}$. The *violation digraph* is the subdigraph $D_{\perp} = (N, A_{\perp})$ of $D_{\leq}$ with $A_{\perp} = \{(i, j) \in N^2 \mid \mathbf{x}_i \leq \mathbf{x}_j \text{ and } y_i > y_j\}$. Their underlying graphs $G_{\leq} = U(D_{\leq}) = (N, E_{\leq})$ and $G_{\perp} = U(D_{\perp}) = (N, E_{\perp})$ are called the *comparability graph* and the *violation graph*, respectively.

The violation graph plays an important role in many monotonic regression algorithms [10, 13, 36] and in monotonicity testing [6]. It is sometimes simply called the *nonmonotonicity graph* [32].

Example 3.2. The comparability/violation (di)graphs associated with the dataset of Example 2.4 may be visualized as follows:

$$\begin{array}{ll} D_{\leq} : & 1 \leftarrow 2 \rightarrow 3 \\ D_{\perp} : & 1 \quad 2 \rightarrow 3 \end{array} \qquad \begin{array}{ll} G_{\leq} : & 1 \text{ --- } 2 \text{ --- } 3 \\ G_{\perp} : & 1 \quad 2 \text{ --- } 3 \end{array}$$

3.3 Monotonic Regression

To say that a dataset $(\mathbf{x}, y)$ is monotone is to say that its labeling $y : N \rightarrow \mathbb{R}$ is a monotone function with respect to the arcs $A_{\leq}$. We let $\mathcal{M}(\mathbf{x})$ denote the set of all monotone labelings:

$$\mathcal{M}(\mathbf{x}) = \{z : N \rightarrow \mathbb{R} \mid \mathbf{x}_i \leq \mathbf{x}_j \text{ implies } z_i \leq z_j\}. \quad (5)$$

Clearly now $(\mathbf{x}, y)$ is monotone if and only if $y \in \mathcal{M}(\mathbf{x})$. When $y \notin \mathcal{M}(\mathbf{x})$, we can consider a labeling $\hat{y} \in \mathcal{M}(\mathbf{x})$ which is closest to y relative to some definition of distance: that $\hat{y}$ is called the monotonic regression of y . Another way to think of this is as making changes to y in order to make it monotone. More formally:

Definition 3.3. Let $(\mathbf{x}, y)$ be a dataset of observations and ℓ a loss function on pairs of labelings; an ℓ -monotonic regression of y with respect to $\mathbf{x}$ is any monotone labeling $\hat{y}$ that minimizes $\ell(y, -)$:

$$\hat{y} \in \arg \min_{z \in \mathcal{M}(\mathbf{x})} \ell(y, z). \quad (6)$$

Following [35], we define the ℓ -distance of y to monotonicity by $\Delta_{\ell}(y) = \ell(y, \hat{y})$. The *residual* of an observation i (with respect to $\hat{y}$), denoted r_i , is given by the difference $r_i = \hat{y}_i - y_i$.

Note that the choice of ℓ determines whether the minimum described in (6) actually exists. The choices of ℓ that we will discuss in this paper do admit such minima, so that this does not pose a concern. In particular, we will look at the following two choices:

Weighted least squares loss. Definition 3.3 is a generalized version of monotonic regression because it is parameterized by a loss function ℓ . The standard definition uses the particular choice of *weighted sum of squared errors* $L_2(y, z) = \sum_{i=1}^n w_i (y_i - z_i)^2$ for some weight vector $w_1, \dots, w_n$ that allows particular emphasis to be placed on certain observations. Also note that, since the set of observations is finite, a function in $\mathcal{M}(\mathbf{x})$ is just a choice of n real numbers that satisfy the monotonicity constraints. Spelling out (6) with this in mind, we get the usual definition of monotonic regression:

$$\arg \min_{z \in \mathcal{M}(\mathbf{x})} L_2(y, z) = \arg \min_{z_1, \dots, z_n \in \mathbb{R}} \sum_{i=1}^n w_i (y_i - z_i)^2 \quad (7)$$

subject to $z_i \leq z_j$ whenever $\mathbf{x}_i \leq \mathbf{x}_j$.

In the coming sections we will use unit weights $w_i = 1$.

The 0-1 loss function L_0 . The 0-1 loss function L_0 , also known as the *Kronecker delta loss function*, simply counts the number of observations on which the labelings disagree:

$$L_0(y, z) = |\{i \in N \mid y_i \neq z_i\}|. \quad (8)$$

The quantity $L_0(y, z)$ is also referred to as the *Hamming distance*. The L_0 -regression is another frequently encountered form of regression (see e.g. [35]). Note that whereas L_2 -regression is incentivized to make (potentially many) small changes, the L_0 -regression is incentivized to make only a few (potentially large) changes.

3.4 Definitions of Nonmonotonicity Metrics

We now consider a number of (non)monotonicity metrics proposed in the literature on monotonic machine learning. We will show that these metrics can be seen as instances of well-known concepts from graph theory, or as metrics related to (generalized) monotonic regression. This overview is closely related to that of [4]; the aim is to be more comprehensive here, and to give additional graph-theoretic interpretations of the measures. Most of these measures are *global*, in the sense that they quantify the degree of nonmonotonicity of the entire dataset at once. Some can be seen as *local* measures, in the sense that they quantify nonmonotonicity of individual observations or pairs of observations. We will briefly remark on the difference between these, where applicable.

3.4.1 Nonmonotonicity Index One. In the terminology of [25], an observation i *clashes* with an observation j when $\{i, j\}$ is nonmonotone. The number of clashes with i is denoted $n_{\text{clash}}(i)$:

$$n_{\text{clash}}(i) = |\{j \in N \setminus \{i\} \mid \{i, j\} \text{ is nonmonotone}\}|. \quad (9)$$

The first nonmonotonicity index NMI_1 is now defined by [25] as:

$$\text{NMI}_1 = \frac{1}{n(n-1)} \sum_{i=1}^n n_{\text{clash}}(i). \quad (10)$$

There are two ways in which this quantity can be expressed in graph-theoretic terms. To start, we note that $n_{\text{clash}}(i)$ is just the degree of i in the violation graph $G_\perp$. Therefore, by the *degree sum formula*, we have $\sum_{i=1}^n n_{\text{clash}}(i) = \sum_{i=1}^n \deg(i) = 2|E_\perp|$, and so:

$$\text{NMI}_1 = \frac{2|E_\perp|}{n(n-1)} = \frac{\deg(G_\perp)}{n-1}. \quad (11)$$

In other words, the NMI_1 can be seen as the normalized average degree of $G_\perp$. Note that the degree $\deg(i)$ of an individual observation i can itself be seen as a local measure of nonmonotonicity.

Viewed differently, the NMI_1 is just the edge density of $G_\perp$:

$$\text{NMI}_1 = \frac{2|E_\perp|}{n(n-1)} = \frac{|E_\perp|}{\binom{n}{2}} = \rho(G_\perp). \quad (12)$$

This means we can intuitively think of the NMI_1 as the probability that an arbitrary (unordered) pair of observations is nonmonotone.

3.4.2 Nonmonotonicity Index Two. The second index, the NMI_2 , is defined by [25] in terms of the rate at which observations clash with one another. More specifically, it is defined in [25] as:

$$\text{NMI}_2 = \frac{1}{n} |\{i \in N \mid n_{\text{clash}}(i) > 0\}|. \quad (13)$$

Note that $n_{\text{clash}}(i) = \deg(i) > 0$ holds just when i is a non-isolated vertex in $G_\perp$. Writing $i(G_\perp)$ for the number of isolated vertices in $G_\perp$, we have $|\{i \in N \mid n_{\text{clash}}(i) > 0\}| = n - i(G_\perp)$ and so:

$$\text{NMI}_2 = 1 - \frac{i(G_\perp)}{n}. \quad (14)$$

This is to say that the NMI_2 is the normalized number of non-isolated vertices in $G_\perp$. This can be thought of as a global measure of non-monotonicity.

The NMI_2 , or rather the quantity $i(G_\perp)/n$, is what is considered as the relative frequency of consistent cases by e.g. [31, 39]. It is stated in [4] that “[the NMI_2] has not been used in any study yet,” though a few months later [5] appeared in which the score was reported for several datasets. Later works such as [31, 39] report (one-minus) NMI_2 scores but do not refer to them as such.

3.4.3 Nonmonotonicity Index Three. The third monotonicity index is defined in [25] as the number of output values in the dataset that would need to be changed in order to make the dataset monotone, divided by the total number of examples n . As mentioned in [25], this quantity cannot reach the value 1, because there is always at least one label that does not need to be changed. The NMI_3 may be understood as the (normalized) L_0 -distance to monotonicity that we discussed in Section 3.3:

$$\text{NMI}_3 = \frac{\Delta_{L_0}(y)}{n}. \quad (15)$$

Again, this metric admits an interpretation in terms of a well known graph-theoretic concept. A *vertex cover* of a graph $G = (V, E)$ is a subset of vertices V' with the property that $\{u, v\} \in E$ implies $u \in V'$ or $v \in V'$. The *vertex cover number* of G , denoted $\tau(G)$, is the cardinality of a smallest vertex cover of G . It is well known that $\Delta_{L_0}(y) = \tau(G_\perp)$; see e.g. [14, Corollary 2] and [10, Section 3]. This means that the NMI_3 can be seen as the normalized vertex cover number of the violation graph $G_\perp$.

It seems that the NMI_3 has not been used often—or even at all—and in fact [4] does not list it in its overview. However, [31] does compute the “number of deletions required” to attain consistency (i.e., monotonicity). This is done by successively deleting the case causing the most inconsistency. This quantity can be seen as an approximation of the NMI_3 value [23].

Lastly, we should mention that computing the vertex cover number of a graph is NP-hard in general, and therefore not feasible to do for large graphs. However, the arc relation of the violation digraph $D_\perp$ is a strict partial order, which means that the vertex cover number of $G_\perp$, and as such the NMI_3 of a dataset, can be computed in polynomial time [16].

3.4.4 (Normalized) ℓ -distance to Monotonicity. The NMI_3 can thus be seen as an instance of the more general idea of using the ℓ -distance to monotonicity of y as a measure of nonmonotonicity for the dataset. The NMI_3 uses $\ell = L_0$, but we could also take the L_2 function discussed in Section 3.3, or any one of the other many available choices. Interestingly, the L_2 -monotonic regression has been used for this exact purpose in the literature on genetics: [15] writes that “[the L_2 -distance to monotonicity] quantifies the monotonicity of a particular [genotype-phenotype] map in terms of the least-squares distance to the closest monotone map.”

Note also that, for an observation i , the residual $r_i = \hat{y}_i - y_i$ with respect to the ℓ -monotonic regression can also be used as a local measure of nonmonotonicity. For this purpose the L_2 loss is preferable to the L_0 loss. This is because the L_2 -regression admits a unique solution, so each residual r_i is uniquely defined. By contrast, the L_0 -regression is not necessarily unique, and the value of r_i may depend heavily on the chosen solution. Take for example the dataset $\{(0, 1), (1, 0)\}$ with L_0 -regression solutions $\hat{y} = (0, 0)$ and $\hat{y}' = (1, k)$ for arbitrarily large $k \geq 1$, then $\hat{y}_2 - y_2 = 0$ while $\hat{y}'_2 - y_2 = k$.

3.4.5 Conditional Edge Density. It is remarked in [25] that a variant of the NMI_1 is considered by [21], which does not divide by the total number of possible edges, but instead by the number of comparable pairs of observations. This measure is referred to by [21] as the *degree of monotonicity*, and is denoted by δ . This same measure is proposed under the same name in [9], and is also used in [41].

Since “degree of monotonicity” is somewhat nondescript in the context of the present work, we will instead refer to this measure as the *conditional edge density* (CED) and define it as:

$$\text{CED} = \frac{|E_{\perp}|}{|E_{\leq}|}. \quad (16)$$

We can intuitively think of the CED as the probability that an arbitrary (unordered) pair of observations is nonmonotone, conditioned on the fact that the pair is comparable. It is thus a more strict version of the NMI_1 ; in fact, the NMI_1 is equal to the CED scaled by the edge density of the comparability graph $G_{\leq}$:

$$\rho(G_{\leq}) \cdot \text{CED} = \frac{|E_{\leq}|}{\binom{n}{2}} \cdot \frac{|E_{\perp}|}{|E_{\leq}|} = \frac{|E_{\perp}|}{\binom{n}{2}} = \text{NMI}_1. \quad (17)$$

3.4.6 Goodman and Kruskal’s γ . Goodman and Kruskal’s γ value is also mentioned as a monotonicity measure in [4], an idea attributed to [24]. It is defined in terms of the quantities $S_+ = |E_{\leq}| - |E_{\perp}|$, which is the number of monotone pairs, and $S_- = |E_{\perp}|$, which is the number of nonmonotone pairs. Now γ is defined as:

$$\gamma = \frac{S_+ - S_-}{S_+ + S_-} = \frac{|E_{\leq}| - |E_{\perp}| - |E_{\perp}|}{|E_{\leq}| - |E_{\perp}| + |E_{\perp}|} = \frac{|E_{\leq}| - 2|E_{\perp}|}{|E_{\leq}|} = 1 - 2\text{CED}. \quad (18)$$

In other words, γ scales the CED by 2, and then shifts and flips it to the $[-1, 1]$ interval. As such, it is equally informative as the CED.

3.4.7 Violation Weight. When a pair of observations (i, j) is non-monotone then that means that $y_i > y_j$ holds rather than $y_i \leq y_j$. We can thus consider the quantity $|y_i - y_j|$ to give an indication of the severity of the violation. This quantity is considered in [6] and is referred to as the *violation weight*. It is a local measure of nonmonotonicity that is focused on pairs of observations rather than on individual observations. It can be made global simply by summing over the weights, and we will denote this quantity by vw :

$$\text{vw} = \sum_{\{i,j\} \in E_{\perp}} |y_i - y_j|. \quad (19)$$

3.5 Monotonicity-Metric Interrelations

Little if any seems to have been written about how the various metrics considered in Section 3.4 interrelate. We will look at a few relations here, in order to inform our data analyses in the

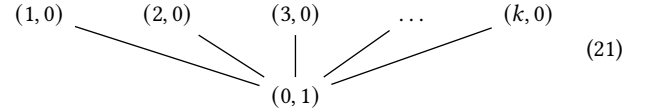
next section. Of course, much more could be said, but a complete comparison between the measures is beyond the scope of this work.

To start, we note that there are datasets that have an NMI_1 score arbitrarily close to 0, but an NMI_2 score arbitrarily close to 1.

Example 3.4. Consider a scenario with one-dimensional input vectors, so that the $\mathbf{x}_i$ values are just singular real numbers. We define a sequence of observation datasets $N^{(k)}$ for $k \geq 1$ as follows:

$$N^{(k)} = \{(0, 1)\} \cup \{(l, 0) \mid 1 \leq l \leq k\}. \quad (20)$$

The violation graph $G_{\perp}^{(k)}$ on $N^{(k)}$ then looks like:



We thus have $|N^{(k)}| = k + 1$ and $|E_{\perp}^{(k)}| = k$, and so:

$$\text{NMI}_1(N^{(k)}) = \frac{k}{\binom{k+1}{2}} = \frac{2}{k+1}. \quad (22)$$

This means $\lim_{k \rightarrow \infty} \text{NMI}_1(N^{(k)}) = 0$. However, $i(G_{\perp}^{(k)}) = 0$, and so

$$\text{NMI}_2(N^{(k)}) = 1 - \frac{0}{k+1} = 1. \quad (23)$$

To summarize: for large enough k we have $\text{NMI}_1(N^{(k)}) \approx 0$ while $\text{NMI}_2(N^{(k)}) = 1$, which is to say that $N^{(k)}$ is practically monotone in one sense and practically nonmonotone in another.

Example 3.4 shows us that the NMI_2 of a dataset can be high while its NMI_1 is low. What about the converse situation, in which a dataset has a high NMI_1 value but a low NMI_2 value? In fact this cannot happen, because the NMI_1 score is capped by the NMI_2 score:

PROPOSITION 3.5. *For every dataset, $\text{NMI}_1 \leq \text{NMI}_2^2 \leq \text{NMI}_2$.*

PROOF. Let N be some dataset and $m = n - i(G_{\perp})$, so $\text{NMI}_2 = m/n$. Every edge in $G_{\perp}$ connects only to nonisolated vertices, so $G_{\perp}$ has at most $\binom{m}{2}$ edges, meaning $|E_{\perp}| \leq \binom{m}{2}$. Therefore:

$$\text{NMI}_1 = \frac{|E_{\perp}|}{\binom{n}{2}} \leq \frac{\binom{m}{2}}{\binom{n}{2}} = \frac{m}{n} \cdot \frac{m-1}{n-1} \leq \left(\frac{m}{n}\right)^2 = \text{NMI}_2^2, \quad (24)$$

where the last inequality follows from $m \leq n$. Furthermore, it follows from $0 \leq \text{NMI}_2 \leq 1$ that $\text{NMI}_2^2 \leq \text{NMI}_2$. $\square$

The ratio of the NMI_1 to the NMI_2 can itself be informative: when the NMI_1 is low but the NMI_2 is high, then this means that there are relatively few edges that nevertheless connect to many vertices. A possible cause of this is that a few vertices have a very high degree. This is showcased by Example 3.4, in which a single observation accounts for all of the nonmonotonicity, and indeed the value of $\text{NMI}_1(N^{(k)})/\text{NMI}_2(N^{(k)}) \approx 0$ for large k .

The NMI_3 is another way of quantifying how “spread out” the nonmonotonicity is. Indeed, reconsidering Example 3.4, we see that since $\{(0, 1)\}$ is a minimum vertex cover of $G_{\perp}^{(k)}$, we have that $\text{NMI}_3(N^{(k)}) = 1/(k+1)$ and so $\lim_{k \rightarrow \infty} \text{NMI}_3(N^{(k)}) = 0$. More generally, as with the NMI_1 , the NMI_3 cannot exceed the NMI_2 :

PROPOSITION 3.6. *For every dataset, $\text{NMI}_3 \leq \text{NMI}_2$.*

PROOF. We have $\tau(G_{\perp}) \leq n - i(G_{\perp})$ as the nonisolated vertices are a vertex cover, so $\text{NMI}_3 = \tau(G_{\perp})/n \leq (n - i(G_{\perp}))/n = \text{NMI}_2$. $\square$

Table 1: Overview of the (non)monotonicity metrics discussed in Section 3 that are used in the data analyses in Section 4. The first five are global, and the last two are local measures.

Name	Notation	Definition
Nonmonotonicity index one	NMI_1	$ E_{\perp} /\binom{n}{2}$
Nonmonotonicity index two	NMI_2	$1 - i(G_{\perp})/n$
Nonmonotonicity index three	NMI_3	$\Delta_{L_0}(y)/n$
Conditional edge density	CED	$ E_{\perp} / E_{\leq} $
Sum of violation weights	VW	$\sum_{\{i,j\} \in E_{\perp}} y_i - y_j $
Vertex degree	$\deg(i)$	$ \{j \mid \{i, j\} \in E_{\perp}\} $
L_2 -regression residual	r_i	$\hat{y}_i - y_i$

4 Data Analyses

In this section, we apply the measures from Section 3 to two legal datasets: COMPAS recidivism scores [2, 34] and Pennsylvania bail decisions [42]. The goal with these analyses is two-fold: on the one hand, it is to assess the decision monotonicity of these datasets, and on the other to evaluate the measures’ differences and usefulness.

We analyze COMPAS scores in Section 4.1 and bail decisions in Section 4.2, using five global and two local measures of monotonicity; an overview is given in Table 1. For both datasets we first look at overall monotonicity in terms of the global measures, and then consider specific cases as identified by the local measures. The results for the global measures are given in Table 2. Tables 3 and 4 respectively describe the analysis results for the local measures.¹

4.1 COMPAS Risk Scores

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) program is proprietary software by Equivant (formerly Northpointe) used in the US to predict outcomes such as recidivism. After the bias allegations of [2], it has become a standard case study in debates on algorithmic fairness. An extended version of the data was published in [34] and was the subject of a case-base consistency analysis in [39]. Of the measures listed in Table 1, [39] measured only $1 - NMI_2 = i(G_{\perp})/n$ scores, and found these values to be very close to 0, i.e. $NMI_2 \approx 1$. Furthermore, [39] did not consider whether there are any specific cases in the dataset that are responsible for this low consistency percentage. As such, this is a good use case to see whether the other measures proposed in Section 3 can offer a more complete picture of the situation.

4.1.1 Contents of the Dataset. Each row in the data describes a COMPAS-scored defendant, with raw and decile scores for (i) general recidivism, (ii) violent recidivism, and (iii) failure to appear. Raw scores are the direct outputs of the COMPAS program. They are proprietary and difficult to interpret. Decile scores are computed from the raw scores on the basis of norm groups and are easier to interpret as they lie between 1 (lowest) and 10 (highest). See [39, Section 2] for a more detailed explanation of the scores. We assume negative correlation with risk for the current age, age at first arrest, marital status, and misdemeanor features, and positive correlations for the others. After filtering, the dataset contains 8985 rows.

¹Code is available at <https://github.com/wijnandmpi/casebase-consistency-metrics>.

Table 2: Four global nonmonotonicity scores for the three outputs of COMPAS, based on the dataset of [34], and of the Pennsylvania bail dataset of [42]. The metrics are computed with respect to the raw outputs of COMPAS, and the amount of bail set in the bail cases. Scores are rounded to two decimals.

Dataset	NMI_1	NMI_2	NMI_3	CED
General Recidivism	0.03	0.98	0.57	0.11
Violent Recidivism	0.02	0.99	0.58	0.08
Failure to Appear	0.08	1.00	0.62	0.34
Bail Amount	0.00	0.98	0.43	0.20

4.1.2 Global Monotonicity Scores. Table 2 reports the global scores for the raw COMPAS outputs using the feature set from [2, 34]. All three scores are reminiscent of Example 3.4, in the sense that they have $NMI_1 \approx 0$ but $NMI_2 \approx 1$. Concretely, this means that while practically every case conflicts with some other case, the vast majority of decision pairs are consistent. Even when we consider the CED, which is a stricter version of the NMI_1 , the values for the general and violent recidivism scores are ≈ 0.1 : for those scores, COMPAS follows the precedent that it has set for itself in 90% of case pairs.

4.1.3 Local Monotonicity Scores. Next we consider the violation degrees and the L_2 -regression residuals of individual score assignments. Since the residuals would be very difficult to interpret for the raw scores, we use the deciles instead. However, this means that we can only use the violent recidivism risk score, because the others were produced using multiple norm groups [39, Table 3]. The residuals are computed using the algorithm of [13].

The individual and joint distributions of the degrees and residuals are visualized in Figure 1. These plots show that, according to the local measures, the majority of cases are quite monotone. Around 60% of cases have not had their score changed at all (indeed, the $NMI_3 \approx 0.3$ for the risk-of-violence deciles), and a further 30% have had their score adjusted by just 1. Similarly, we can see that the distribution of the degrees is extremely skewed; a very small number of cases have disproportionately high degrees. The highest-degree case i has $\deg(i) = 4039$, meaning that it alone produces a lower bound of $NMI_2 \geq (4039 + 1)/8985 \approx 0.45$. Lastly, we note that the joint distribution shows that the degrees and residuals really measure two different things: a case can have a very high degree but a low residual, and vice versa. The Pearson correlation coefficient between the absolute residuals and the degrees is roughly 0.57.

Finally, we look at the feature values of some specific cases that have particularly high degrees or absolute residuals: see Table 3. Since the COMPAS data contains too many input features to list in the table, we report only the (reconstructed) subscale values indicating criminal involvement, violent history, and history of noncompliance. We also compare these cases to [34, Table 6], which identifies a number of persons who received low COMPAS risk scores despite presenting extensive criminal records. Intuitively, these are exactly the type of cases that one would expect to turn up as being highly locally inconsistent, and so it is interesting to check how they compare to the cases identified by our measures.

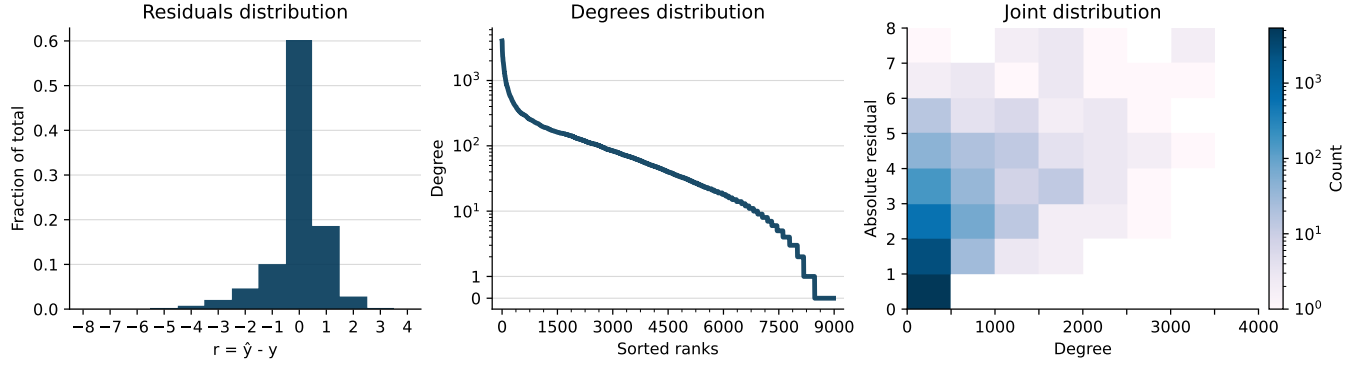


Figure 1: L_2 -residual and degree distributions for the violent recidivism decile scores in the COMPAS dataset of [34]. The vast majority of cases have relatively low degree ($\deg(i) < 100$) and a small residual ($-1 \leq r_i \leq 1$), so log scales are used.

Table 3: Notable cases in the COMPAS dataset, with selected rows from [34, Table 6]. Here y is the violent recidivism decile score, $\hat{y}$ its L_2 -monotonic regression, and we show three reconstructed subscales: criminal involvement (crinv), violent history (vihst), and history of noncompliance (hstnc).

i	crinv	vihst	hstnc	y_i	$\hat{y}_i$	r_i	$\deg(i)$
7445	18	3	6	1	3	2	107
857	27	3	7	1	2	1	81
1295	20	6	6	3	4	1	41
2123	22	5	2	3	3	0	6
5829	32	5	10	2	2	0	4
7294	31	5	1	3	3	0	1
5498	63	10	10	3	7	4	100
6497	62	6	12	1	5	4	188
6607	37	5	10	2	6	4	203
8407	0	0	0	5	1	-4	4039
5965	0	0	0	9	1	-8	4016
3053	0	0	0	9	1	-8	3243
2562	0	0	0	10	2	-8	3195

The first two blocks of Table 3 make up the top-3 and bottom-3 rows of [34, Table 6] in terms of degrees. These rows have relatively low degrees and residuals, meaning that from a monotonicity perspective they are not that extraordinary. The third block shows the cases with the overall highest residuals. These have substantially higher subscale scores, as well as higher degrees than those of [34, Table 6], while also having received low scores from COMPAS. This suggests that the L_2 -residuals provide a general method of finding decision outliers in settings where the monotonicity assumption applies. Block four shows the rows with the overall highest degrees, which happen to coincide with the rows that have the largest negative residuals. These present minimal criminal records, but very high risk assessment scores. The last row of the table was assigned the highest possible violent recidivism decile score, while scoring zeros on the subscales. It participates in a total of 35 nonmonotone pairs with the highest attainable violation weight.

4.2 Pennsylvania Bail Decisions

Our second case study uses the Pennsylvania Bail Decision dataset of [42]. As with COMPAS, we first report global monotonicity measures and then use local measures to look at specific cases. An interesting aspect of this dataset is that the decisions are made by different magistrates, and that these are identified as such in the data. As mentioned in Section 3.1, nonmonotonicity can arise by grouping decisions of individually monotone decisionmakers. As a proof of concept, we use a statistical test to see if the magistrates are more or less monotone individually than they are together.

4.2.1 Contents of the Dataset. In [42] docket and criminal-history information were collected from the Unified Judicial System of Pennsylvania Web Portal, yielding roughly 90,000 cases from 2016–2019. Each row includes defendant characteristics, criminal history, charge information, and bail decision. Some likely relevant factors are missing (e.g., finances), so zip-code proxies were added (income, education, employment). See [42, Section 4] for further details. We assume all features positively correlate with bail amount, except age and the zip-code proxies, and only consider cases where a monetary bail amount was set. After filtering, about 32,000 cases remain.

4.2.2 Global Monotonicity Scores. The last row of Table 2 reports NMI_1 , NMI_2 , NMI_3 , and CED for bail amounts. The pattern matches COMPAS: very small NMI_1 but very large NMI_2 . The degree and residuals distributions are equally similar and so, for the sake of brevity, we will not list them here. Instead, we turn to local measures and review some individually noteworthy cases.

4.2.3 Local Monotonicity Scores. As before, Table 4 reports the residuals of the L_2 -monotonic regression alongside violation graph degrees. The results again show that residual size and degree are largely independent: the largest negative residual has degree only two, and the Pearson correlation between absolute residuals and degrees is just 0.16. The largest positive residuals correspond to cases with many charges and severe lead offenses that nevertheless received relatively low bail. Conversely, the highest bail amounts, including an outlier of \$15,000,000 relative to a median of \$20,000, produce little or no inconsistency; the two largest bail amounts have degree zero. The highest-degree cases instead arise in densely

Table 4: Notable cases in the Pennsylvania bail dataset by L_2 -monotonic regression residuals and degrees. Bail quantities ($y_i, \hat{y}_i, r_i$) are in USD millions. Of the features used to compute monotonicity we only list lead offense (loff), number of charges (nchrg), and number of priors (nprr).

i	loff	nchrg	nprr	y_i	$\hat{y}_i$	r_i	deg(i)
15244	F1	14	0	10.00	3.73	-6.27	2
17575	F3	1	0	5.00	0.03	-4.97	9726
20750	F3	1	8	5.00	0.03	-4.97	1295
12321	F1	89	0	0.10	5.00	4.90	580
4066	F1	39	0	0.45	3.73	3.28	188
7494	F1	28	1	0.75	3.73	2.98	29
12127	F1	10	4	15.00	15.00	0.00	0
15239	F1	15	0	10.00	10.00	0.00	0
15244	F1	14	0	10.00	3.73	-6.27	2
17575	F3	1	0	5.00	0.03	-4.97	9726
18026	M1	1	0	0.10	0.01	-0.09	7668
5239	M1	1	0	0.75	0.01	-0.74	7577

populated regions of the input space, where even small bail discrepancies can induce substantial nonmonotonicity. This contrasts with the intuitive expectation that extreme high-bail outliers would be the main source of inconsistency.

4.2.4 Collective vs. Individual Monotonicity. Lastly, we briefly consider whether magistrates are individually more monotone than they are together. A total of 461 magistrates appear in the filtered data, though only 229 have at least 10 decisions, and 25 have at least 200. To check if the observed assignment of magistrates has an effect on monotonicity we use permutation tests [12]: we repeatedly randomize the assignment of cases to magistrates, recompute a magistrate-dependent statistic t , and compare it to the observed value t^* . With m permutations and statistics t_i , we estimate

$$\hat{p} = \frac{1 + |\{i \in \{1, \dots, m\} \mid t_i \leq t^*\}|}{1 + m}. \quad (25)$$

Values of $\hat{p}$ close to 0 indicate that the observed statistic is unusually small under the null hypothesis that the assignment does not matter; similarly, values close to 1 are unusually high.

Figure 2 shows the resulting p -value distributions for NMI_1 , NMI_2 , CED, and vw, computed separately for the ≥ 10 and ≥ 200 subsets. Under the null, the p -values should be approximately uniformly distributed on $[0, 1]$. This is broadly what we see for most metrics in both subsets. The main deviation occurs for vw: among the 25 high-volume magistrates, many have $\hat{p} < 0.05$, suggesting magistrate assignment may affect violation severity.

However, we stress that this test is a proof of concept: it operates under the strong assumption that magistrates tend to see similar cases. If some handle systematically milder or more severe cases, then random reassignment produces artefacts that influence the results. For example, a magistrate who usually sees mild cases will tend to set low bail amounts, yielding small violation weights. If that magistrate is later randomly assigned more severe cases, the associated violation weights will tend to increase accordingly.

5 Conclusion and Future Work

This article connects monotonicity constraints in machine learning with precedential constraints in formal legal theory by showing that Horty’s result model is a special case of monotonicity constraints studied in computer science. We organize monotonicity metrics, give graph-theoretic interpretations and relations among them, and illustrate their complementary behavior on COMPAS risk scores and Pennsylvania bail decisions, including the localization of highly inconsistent cases. The resulting framework provides both a mathematical basis and practical tools for case-base consistency analysis.

Several directions remain open. First, we considered only L_0 - and L_2 -regression scores, while other choices of ℓ may yield useful nonmonotonicity metrics. A systematic comparison could clarify which metrics are preferable in which settings. Second, we established relations only among the first three nonmonotonicity indices; further relations could help guide data analyses such as those in Section 4. Finally, it would be useful to formulate a general definition of a monotonicity metric. For example, if $\mathcal{D}$ is the set of possible datasets, a nonmonotonicity measure could be defined as a function $m : \mathcal{D} \rightarrow \mathbb{R}$ satisfying natural desiderata, such as monotonicity with respect to violation graphs: if the violation graph of D is, up to isomorphism, a subgraph of that of E , then $m(D) \leq m(E)$.

Acknowledgments

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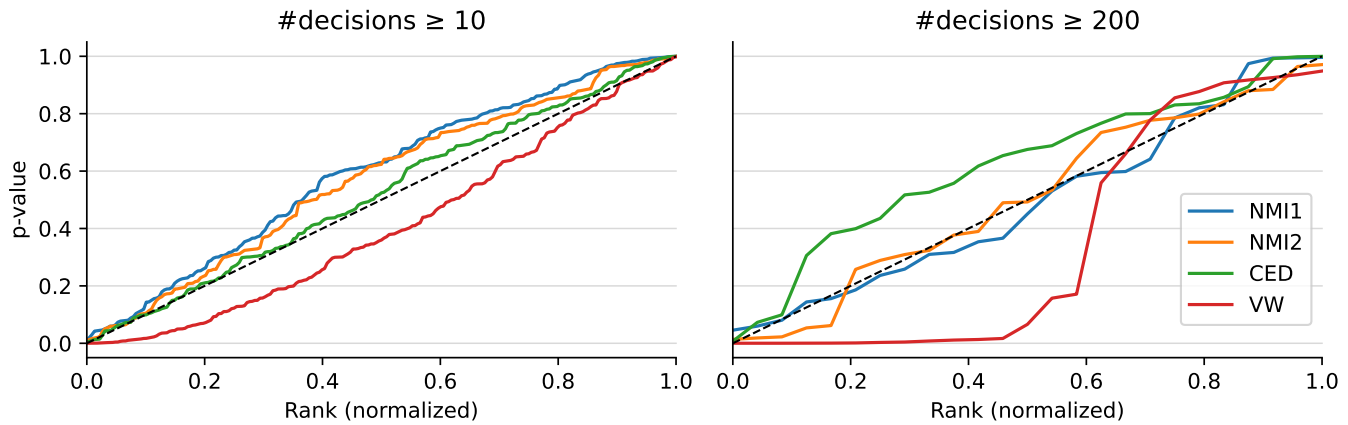


Figure 2: Estimated p -values for NMI_1 , NMI_2 , CED , and VW , for magistrates with ≥ 10 or ≥ 200 decisions, each based on 5,000 permutations. There are 229 magistrates with ≥ 10 decisions and 25 with ≥ 200 .

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